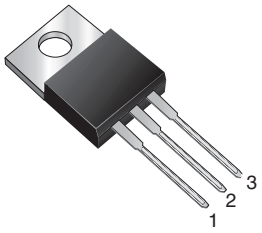
Dual Low-Voltage Trench MOS Barrier Schottky Rectifier

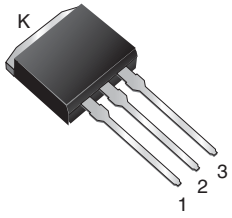
Ultra Low V_F = 0.33 V at I_F = 5.0 A

TMBS®

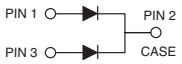
TO-220AB



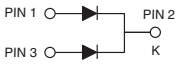
TO-262AA



VT2045C



VIT2045C



FEATURES

- Trench MOS Schottky technology
- Low forward voltage drop, low power losses
- High efficiency operation
- Solder dip 275 °C max. 10 s, per JESD 22-B106
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

RoHS COMPLIANT
HALOGEN FREE

TYPICAL APPLICATIONS

For use in high frequency DC/DC converters, switching power supplies, freewheeling diodes, OR-ing diode, and reverse battery protection.

MECHANICAL DATA

Case: TO-220AB and TO-262AA
Molding compound meets UL 94 V-0 flammability rating Base P/N-M3 - halogen-free, RoHS-compliant, and commercial grade

Terminals: matte tin plated leads, solderable per J-STD-002 and JESD 22-B102
M3 suffix meets JESD 201 class 1A whisker test

Polarity: as marked

Mounting Torque: 10 in-lbs maximum

PRIMARY CHARACTERISTICS	
$I_{F(AV)}$	2 x 10 A
V_{RRM}	45 V
I_{FSM}	160 A
V_F at I_F = 10 A	0.41 V
T_J max.	150 °C
Package	TO-220AB, TO-262AA
Diode variations	Common cathode

MAXIMUM RATINGS (T_A = 25 °C unless otherwise noted)

PARAMETER		SYMBOL	VT2045C	VIT2045C	UNIT
Maximum repetitive peak reverse voltage		V_{RRM}	45		V
Maximum average forward rectified current (fig. 1)	per device	$I_{F(AV)}$	20		A
	per diode		10		
Peak forward surge current 8.3 ms single half sine-wave superimposed on rated load per diode		I_{FSM}	160		A
Operating junction and storage temperature range		T_J, T_{STG}	-40 to +150		°C

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ELECTRICAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)						
PARAMETER	**TEST CONDITIONS**		**SYMBOL**	**TYP.**	**MAX.**	**UNIT**
Instantaneous forward voltage per diode	I_F = 5 A	T_A = 25 °C	V_F (1)	0.44	-	V
	I_F = 10 A			0.49	0.58	
	I_F = 5 A	T_A = 125 °C		0.33	-	
	I_F = 10 A			0.41	0.52	
Reverse current per diode	V_R = 45 V	T_A = 25 °C	I_R (2)	-	2000	µA
		T_A = 125 °C		10	30	mA

Notes

(1) Pulse test: 300 µs pulse width, 1 % duty cycle

(2) Pulse test: Pulse width ≤ 40 ms

THERMAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)					
PARAMETER		**SYMBOL**	**VT2045C**	**VIT2045C**	**UNIT**
Typical thermal resistance	per diode	$R_{\theta JC}$	3.0		°C/W
	per device		2.0		

ORDERING INFORMATION (Example)					
PACKAGE	**PREFERRED P/N**	**UNIT WEIGHT (g)**	**PACKAGE CODE**	**BASE QUANTITY**	**DELIVERY MODE**
TO-220AB	VT2045C-M3/4W	1.88	4W	50/tube	Tube
TO-262AA	VIT2045C-M3/4W	1.45	4W	50/tube	Tube

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RATINGS AND CHARACTERISTICS CURVES (T_A = 25 °C unless otherwise noted)

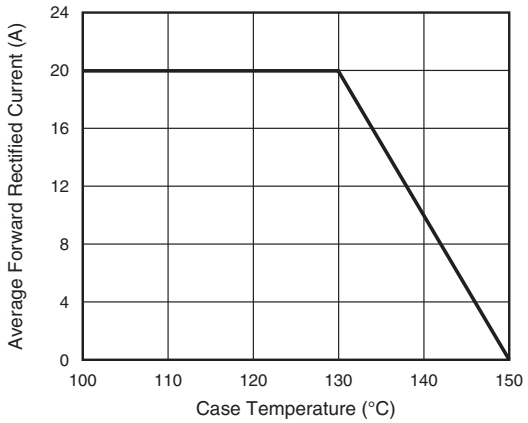


Fig. 1 - Maximum Forward Current Derating Curve

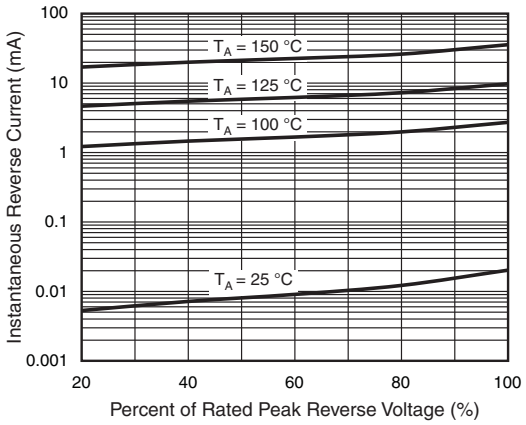


Fig. 4 - Typical Reverse Characteristics Per Diode

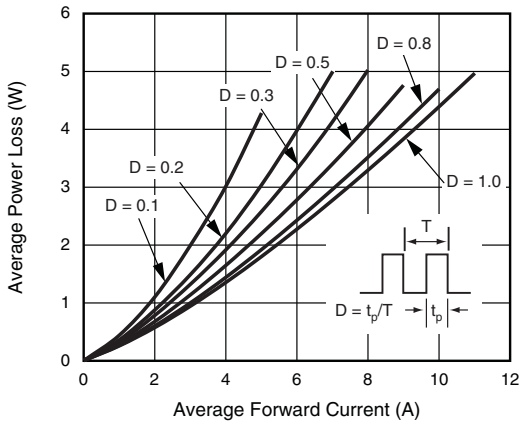


Fig. 2 - Forward Power Loss Characteristics Per Diode

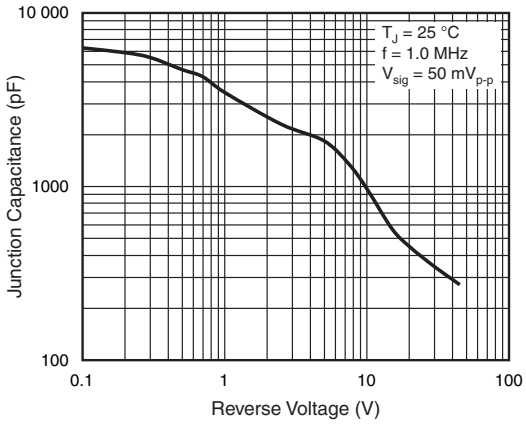


Fig. 5 - Typical Junction Capacitance Per Diode

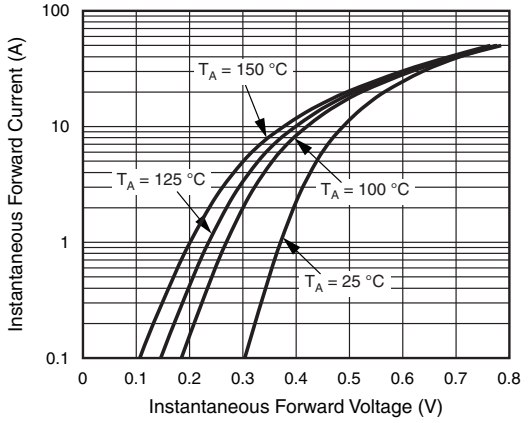


Fig. 3 - Typical Instantaneous Forward Characteristics Per Diode

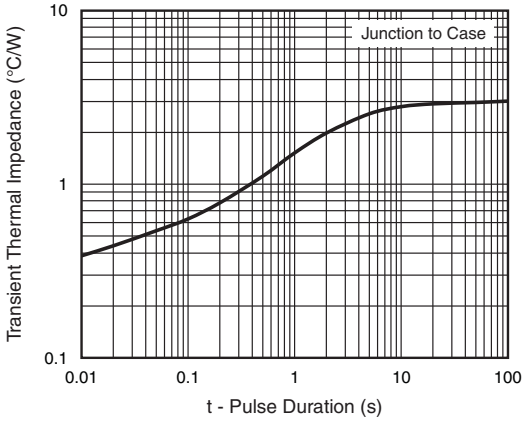


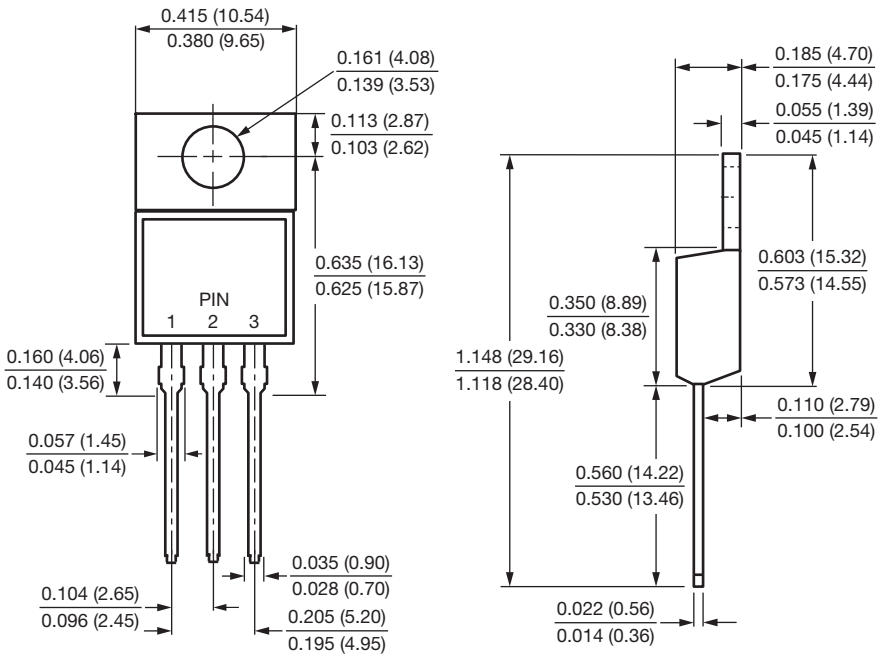
Fig. 6 - Typical Transient Thermal Impedance Per Diode

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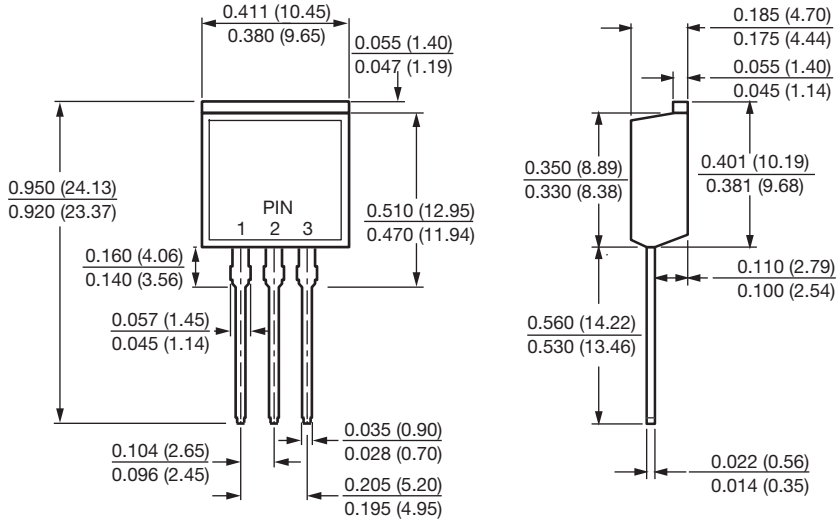
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PACKAGE OUTLINE DIMENSIONS in inches (millimeters)

TO-220AB



TO-262AA



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